

Dataset	Trace substrate	Reliable truth	Paper role
Open-SWE	Real software-agent traces	workflow, repo, owner-like	Main real-data evidence
tau-bench historical	Real tool-agent traces	workflow/session	Non-code external validity
Hierarchy overlay	Real traces + controlled overlay	user / tenant / project	Longitudinal mechanism
Controlled replay	Re-keyed public traces	known workflow structure	Computation scale

Each evidence layer supports a different question: exposure, continuity, hierarchy, or scale.